

CLIMATE CHANGE

The greatest threat to the oceans and coastal communities is climate change. Global warming is melting polar ice, raising sea levels, and making violent oceanic storms more common. Oceans are getting warmer, while carbon dioxide pollution is making the water more acidic. This is destroying coral reefs, and could drive a lot of marine life into extinction. Ocean currents are getting disrupted, and if warmer waters trigger melting of the frozen methane that lies beneath the seabed, the effects could be catastrophic.



▲ THE GREENHOUSE EFFECT

The planet is kept warm by an atmospheric blanket of greenhouse gases that retain some of the heat radiated from its sun-warmed surface. One of the most important of these is carbon dioxide, which can be created by burning coal, oil, and gas. Massive use of these fuels has added more carbon dioxide to the air, increasing the greenhouse effect and causing global warming.



▲ MELTING ICE

Rising temperatures are melting the ice sheets of Antarctica and Greenland. Meltwater streaming off the land is causing a rise in sea level, and making polar oceans less salty. The summer sea ice in the Arctic has also shrunk to the smallest area ever recorded. This threatens Arctic wildlife such as the polar bear, which lives on the ice. If the sea ice disappears, the polar bear could vanish, too.



▲ RISING SEA LEVELS

Water added by melting ice threatens to make average sea levels rise by up to 3 ft (1 meter) over the next century, and possibly more. The effect of this will be exaggerated in many regions that are exposed to large tides and storm surges. Coastal cities such as Shanghai in China could be flooded, low-lying lands such as Bangladesh will be swamped, and several island nations will virtually disappear.